

## René MOURAMBOU DIYEMA

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#) | [Medium](#) | [LeetCode](#) | [HackerRank](#) | Paris, France

I'm an AI engineer with a **Bachelor's** and a **Master's** in **Artificial Intelligence** from **Paris-Saclay University** in **Deep Learning, Computer Vision, NLP and Foudation Models**, seeking a **full-time** role from **October 2026**.

### EDUCATION

**MSc Computer Science, Artificial Intelligence.** Sept 2025 – Sept 2026

Paris-Saclay University, *First French Research University in Mathematics and Artificial Intelligence.*

**MSc Computer Science, Mathematics, Vision, Learning.** Sept 2024 – Sept 2025

ENS Paris-Saclay, *First French Research University in Artificial Intelligence.*

**Msc Computer Science, Artificial Intelligence.** Sept 2023 – Sept 2024

ESIEE Institute of Technology.

**BSc Computer Science, Applied Mathematics.** Sept 2017 – Sept 2019

Sup'Management Engineering School, *First Gabonese Research University in Mathematics.*

### PROFESSIONNAL EXPERIENCE

**Machine Learning Engineer | Internship | Full-Time | Nokia Networks, France:** April 2026 – Sept 2026

- Development of 5G/6G channel estimation methods, using semi-supervised and self-supervised learning.
- Design of CNN-based and Transformer-based architectures for feature extraction and reconstruction.
- Development of end-to-end pipelines with MLflow and Kubeflow for distributed training on GPU clusters.

**Concepts:** Python, PyTorch, TensorFlow, ResNet, Transformers, C++, Signal Processing, CUDA, Git.

**Deep Learning Researcher | Internship | Full-Time | SEGULA Technologies, France:** April 2025 – Sept 2025

- Development of reduced-order models for multi-physics dynamical systems governed by nonlinear PDEs.
- Implementation and fine-tuning of Deep Learning architectures (Autoencoders, LSTM, GNN) for PDEs.

**Concepts:** Python, PyTorch, TensorFlow, LSTM, GNN, C++, Boost, Eigen, Parallel programming, CUDA.

**Machine Learning Engineer | Fixed-Term Contract | Full-Time | Evolukid, France:** Sept 2023 – Sept 2024

- Design of a real-time multi-modal chatbot, with BERT handling intent classification from user input.
- Development of a YOLO-based diagram detection module, with OpenCV preprocessing.
- Integration of a Vision-Language Model using BERT and YOLO outputs for response generation.

**Concepts:** Python, PyTorch, TensorFlow, FastAPI, C++, Text Processing, Image Processing, Docker, Git.

**Software Engineer | Fixed-Term Contract | Part-Time | Fujitsu Technology, France:** Sept 2023 – Sept 2024

- Website backend management, data flow optimization, code optimization and bug correction.
- Search for efficient algorithms for parallel processing of tasks and writing of documentation.

**Concepts:** JavaScript, Angular, PostgreSQL, Parallel Programming, Distributed Systems, AWS.

**Research Scientist | Internship | Part-Time | Applied Artificial Intelligence Lab, France:** Sept 2022 – Aug 2023

- Development of multi-modal system for scientific document understanding.
- Design of CNN architectures for diagram extraction and Transformer models for text representation.

### PROJECTS

**Instruction-Following Multimodal Agent with Vision-Language-Action Reasoning |** Jan 2025 – May 2025

- Built a multimodal agent combining text, vision, and reasoning for instruction following.
- Fused CLIP visual embeddings with open-source LLM (LLaMA) via cross-attention.
- Implemented an "Observation → Reasoning → Action" loop using LangChain.

**Concepts:** Python, PyTorch, C++, CUDA, LangChain, Transformers, Multimodal Learning.

**3D Scene Reconstruction and Estimation using Implicit Neural Representation |** Dec 2023 – Apr 2024

- Dense 3D reconstruction from monocular video using implicit NeRF representations.
- Self-supervised depth and camera pose estimation with photometric consistency.

**Concepts:** Python, Pytorch, PyTorch3D, OpenCV, Open3D, C++, CUDA, NeRF, Differentiable Rendering.

**Unsupervised Morphological and Syntactic structure Induction via VAE |** Nov 2022 – Mar 2023

- Unsupervised learning of linguistic morphology and syntax using a sequence-to-sequence VAE.
- Optimised ELBO to analyse latent representations and linguistic correlations.

**Concepts:** Python, Pytorch, SpaCy, NLTK, C++, CUDA, Bayesian Inference, Statistical Linguistic.

### TECHNOLOGIES AND LANGUAGES

- C++, OpenMP, Boost, Python, PyTorch, TensorFlow, Keras, OpenCV, LLM, VLM, CUDA, Linux, Git, SQL.

Here is a list of the courses I have taken or am currently taking:

**MSc Computer Science, Artificial Intelligence-----Sept 2025 – Sept 2026**

**Paris-Saclay University, France**, *First French Research University in Mathematics and Artificial Intelligence*

- Image Mining and Computer Vision
- Deep Learning and Natural Language Processing
- Automated Speech Recognition and Language Processing
- Text Mining, Chatbot and Large Language Models
- Learning Theory and Advanced Machine Learning
- Scientific Machine Learning and Reinforcement Learning
- Probabilistic Generative Models and Signal Processing
- Advanced Optimization and Automated Machine Learning

**MSc Computer Science, Mathematics, Vision Learning-----Sept 2024 – Sept 2025**

**ENS Paris-Saclay, France**, *First French Research University in Mathematics*

- Statistical and Bayesian Machine Learning
- Probabilistic Graphical Models and Deep Generative Models
- Convex Optimization and Machine Learning Applications
- Large Language Models, and Chatbot
- Algorithms for Speech and Natural Language Processing
- Deep Learning for Image Restoration and Synthesis
- Advanced Deep Learning and Signal Processing
- Object Recognition and Computer Vision

**MSc Computer Science, Mathematics -----Sept 2023 – Sept 2024**

**ESIEE Institute of Technology, France**

- Advanced Calculus and Linear Algebra
- Probability and Statistics
- Discrete Markov Chains
- Stochastic Calculus
- Computer Vision and Image Processing
- Advanced Machine Learning
- Advanced Deep Learning and Natural Language Processing

**BSc Computer Science, Mathematics -----Sept 2017 – Sept 2019**

**Sup'Management Engineering, Gabon**, *First Gabonese Research University in Mathematics*

- Linear algebra and calculus
- Probability and Statistics
- Advanced Python and Advanced C++
- SQL and Graph Theory
- Algorithms and Data Structures